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# Fabrication and Characterization of a pH-Responsive Intelligent Packaging Film From *Amorphophallus paeoniifolius* Starch and *Clitoria ternatea* Extract for Real-Time Chicken Freshness Monitoring

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## ABSTRACT

This study aimed to develop a pH-sensitive intelligent packaging film using starch extracted from elephant foot yam (EFY) (*Amorphophallus paeoniifolius*). The starch was dual-modified and blended with polyvinyl alcohol (PVA) and anthocyanin-rich *Clitoria ternatea* extract (CTE) to produce composite films capable of real-time chicken meat freshness monitoring. LC-MS/MS profiling confirmed the presence of several phenolic and flavonoid compounds, including rutin, delphinidin, and quercetin. Structural analyses using field-emission scanning electron microscopy (FE-SEM), confocal laser scanning microscopy (CLSM), along with spectroscopic techniques such as Fourier transform infrared spectroscopy (FTIR), x-ray diffraction (XRD), indicated good dispersion of CTE within a semicrystalline polymer matrix. Incorporation of CTE modifies film properties: film thickness increased from 0.12 to 0.16 mm, while water vapor permeability decreased from 1.77 to 1.46 g mm/m<sup>2</sup>/day/kPa. Colorimetric sensitivity increased, with  $\Delta E$  values rising from 1.80 to 17.5 across a broad pH range. Mechanical performance was peaked at 3% CTE, resulting in a tensile strength of 16.36 MPa and elongation at break of 24.20%. The films also exhibited notable antioxidant and antimicrobial activity against *Escherichia coli* and *Staphylococcus aureus*. Application to chicken packaging showed distinct color changes correlated with freshness indicators, including pH, microbial growth, TBARS (0.45–2.32 mg MDA/kg), and total volatile basic nitrogen (TVB-N) (10.2–38.7 mg/100 g). Overall, the results suggest that EFY starch-PVA-CTE films are promising sustainable and biodegradable materials suitable for intelligent food packaging and real-time freshness monitoring.

## 1 | Introduction

Food packaging plays a crucial role in the food supply chain by maintaining product integrity, slowing deterioration, and extending shelf-life (Thakur et al. 2024). Petroleum-derived

plastics are widely used in consumer packaging due to their lightweight, visual appeal, ease of processing, affordability, and superior functional properties (Roy and Rhim 2020). However, their extensive use has raised ecological concerns because of poor degradability resulting in long-term environmental accumulation

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of non-biodegradable waste (Jiang et al. 2023; Manzoor et al. 2023). Although bio-based biodegradable polymers are being increasingly explored across various industries, their overall impact on their large-scale adoption remains limited due to higher cost, processing constraints, and variability (Ahmed et al. 2018).

Starch is an abundant, inexpensive, and biodegradable thermoplastic biopolymer composed of amylose and amylopectin (Shanmathy et al. 2021). Its biodegradability and strong film-forming ability make it a promising candidate for food packaging applications (Thakur et al. 2024). Intelligent packaging systems are gaining prominence for enhancing product quality, ensuring food safety, minimizing waste, extending shelf life, and promoting environmental sustainability (Wu et al. 2024). These systems typically incorporate sensors, indicators, or colorimetric agents to enable real-time monitoring of food freshness and quality (Thakur et al. 2024). The global market for bio-based intelligent food packaging is projected to reach \$26.7 billion by 2024 and \$31.98 billion by 2029 (Kumar et al. 2024).

Natural bioactive compounds are increasingly incorporated into biopolymeric films to impart antioxidant and antimicrobial activity. Among these, anthocyanins are widely used as pH-sensitive colorimetric indicators to detect spoilage-related pH variations in food (Jiang et al. 2023). Anthocyanins, possess a three-ring flaviumstructure, exist in multiple structural forms and undergo reversible color transitions across different pH levels. These pH-induced chromatic transitions make anthocyanins ideal for visually monitoring biochemical changes associated with food spoilage (Cheng et al. 2022). *Clitoria ternatea* (butterfly pea flower) is particularly rich in acylated anthocyanins known as ternatins, which exhibit superior stability compared with non-acylated forms (Rashad et al. 2025). These properties make *C. ternatea* extract (CTE) a promising natural dye for intelligent packaging applications. Recent studies have demonstrated the successful incorporation of butterfly pea flower extract into starch-based films, such as avocado seed starch films used as pH-sensitive indicators (Asikkutlu and Yildirim-Yalcin 2025).

Starch from various botanical sources has been used for film fabrication, including maize (Fitch-Vargas et al. 2016), wheat (Basiak et al. 2015), taro (Shanmathy et al. 2021), cassava (Parra et al. 2004), oat (Galdeano et al. 2009), and sweet potato starches (Choi et al. 2022), each exhibiting distinct morphological, rheological, thermal, and functional properties (Sigdel and Janaswamy 2024). Elephant foot yam (EFY) (*Amorphophallus paeoniifolius*) is a tropical tuber commonly consumed in Asia, and its starch has a high extraction yield of 87.7% with an amylose content ranging from 24.2% to 28.29% (Reddy et al. 2014). Its physiochemical and functional characteristics are comparable to those of commercial starches, making it a viable alternative for food and industrial applications (Suriya et al. 2019). Despite these advantages, EFY starch (EFYS) has seen limited use in intelligent films. Native starch-based films often suffer from brittleness, poor mechanical strength, low water resistance, and stickiness, restricting their suitability for commercial application (Singh et al. 2023).

To overcome these limitations, starch is often blended with compatible biopolymers such as polyvinyl alcohol (PVA). PVA

is a biodegradable, nontoxic, and biocompatible polymer, with excellent film-forming and barrier properties, making it suitable for food packaging (Li et al. 2024). Blending starch with PVA improves mechanical strength, flexibility, and stability (Nagar et al. 2020; Arifin et al. 2025). Recent studies have reported starch-PVA-based intelligent packaging films, including jackfruit seed starch/PVA films incorporating purple sweet potato anthocyanins and starch-based films integrating colorimetric indicators for meat freshness monitoring (Nguyen et al. 2025).

The present study developed a pH-sensitive intelligent packaging film by incorporating anthocyanin-rich CTE into dual-modified EFYS-PVA matrices. The aim was to produce a film with enhanced mechanical, barrier, and thermal properties while functioning as a reliable colorimetric freshness indicator.

## 2 | Materials and Methods

### 2.1 | Materials

EFY (*A. paeoniifolius*) tubers and butterfly pea flowers (*C. ternatea*) were procured from a local farmer's market in Visakhapatnam, India. Fresh chicken samples were sourced from a local poultry farm in the same region. All chemicals and reagents used in this study, including glycerol, ethanol, glacial acetic acid, and PVA, were of analytical grade and purchased from Hi-Media Co., Mumbai, India.

### 2.2 | Isolation of Starch and Preparation of Dual-Modified Starch

Starch was extracted following the method described by Reddy et al. (2014). Briefly, EFY tubers were peeled, chopped, and treated with 0.1% (w/v) sodium metabisulfite. The homogenized slurry was suspended in 4% (w/v) NaCl, filtered, and centrifuged. The resulting starch sediment was dried at 45°C for 24 h and stored in airtight containers.

Dual-modification of EFY starch was performed following the method described by Prakruthi et al. (2025). A starch slurry (25% w/v) was ultrasonicated at 40°C for 15 min, centrifuged at 5000 rpm, and oven dried at 45°C to approximately 10% moisture content. The dried starch was then treated with acetic acid (1:4 w/v), incubated at 45°C for 2 h, centrifuged, and dried again. The dual-modified EFY starch (MEFYS) obtained was stored in airtight containers for further use.

#### 2.2.1 | Total Starch Content, Amylose Content, and Gelatinization Temperature

The amylose and total starch contents were determined using the iodine-binding method described by Prakruthi et al. (2025). Thermal properties of the starch, including onset temperature ( $T_o$ ), peak temperature ( $T_p$ ), conclusion temperature ( $T_c$ ), and gelatinization enthalpy ( $\Delta H$ ), were analyzed using a differential scanning calorimeter (DSC-60 Plus, Shimadzu, Japan) over a temperature range of 30–200°C.

## 2.2.2 | Starch Particle Size and Morphology

The particle size of dual-modified EFY starch was measured using a laser diffraction particle size analyzer. For morphological analysis, starch samples were dispersed in ultrapure water, dried, and examined using field-emission scanning electron microscopy (FE-SEM; Tescan Mira-3, Czech Republic). Prior to imaging, the samples were sputter-coated with a thin layer of gold to enhance surface conductivity.

## 2.3 | Extraction of Polyphenols From *Clitoria ternatea* Flower

Polyphenols were extracted from *C. ternatea* flowers following the method described by Netravati et al. (2022) with minor modifications. Fresh petals were cleaned, dried, and ground into a fine powder. The powder was extracted with 50% ethanol containing 1% citric acid (1:10 w/v). The mixture was incubated at 50°C for 12 h, filtered, and concentrated at 50°C using a rotary vacuum evaporator. The resulting blue polyphenol-rich extract was freeze-dried and stored at 4°C for further analysis.

### 2.3.1 | Total Anthocyanin Content (TAC) of CTE

The TAC was determined using a pH differential method as described by Netravati et al. (2022). The concentrated pigment extract was diluted in pH 1.0 and pH 4.5 buffer solutions, prepared using potassium chloride and sodium acetate, respectively. The solutions were incubated for 15 min before absorbance measurements were taken using a spectrophotometer (UH5300 UV-Vis spectrophotometer, Hitachi, Tokyo, Japan). Absorbance was recorded at 530 and 700 nm and was calculated using the following equations:

$$A = (A_{530 \text{ nm}} - A_{700 \text{ nm}})_{\text{pH } 1.0} - (A_{530 \text{ nm}} - A_{700 \text{ nm}})_{\text{pH } 4.5} \quad (1)$$

$$\text{TAC (mg/L)} = \frac{A \times \text{MW} \times \text{DF} \times 1000}{\epsilon \times l} \quad (2)$$

where MW is the molecular weight (449.2 g/mol) of cyanidin-3-glucoside, DF is the dilution factor,  $\epsilon$  is the molar absorptivity coefficient (29,600 L/mol/cm), and  $l$  is the optical path length (1 cm).

### 2.3.2 | LC-MS/MS (QTRAP) Analysis of CTE

Metabolite profiling of the extract was performed using a liquid chromatography system coupled with electrospray ionization tandem mass spectrometry (LC-ESI-MS/MS) on a SCIEX QTRAP 5500+ system. The extract was dissolved in MS-grade methanol and stored in glass vials prior to analysis. Chromatographic separation was achieved using a Phenomenex C18 column (150 × 3 mm<sup>2</sup>, 2.6 μm, 100 Å), operated with a binary pump system. The operating conditions followed the method of Reddy et al. (2020).

## 2.4 | Development of pH-Intelligent Films

The composite films containing PVA, EFYS, and CTE anthocyanins were prepared following the method outlined by Li et al. (2022), with slight modifications. EFYS and PVA were separately gelatinized and dissolved to form 10% (w/v) solutions by heating at 80°C for 30 min and 100°C for 1 h, respectively. The two solutions were then mixed in a 60:40 ratio. Glycerol and glacial acetic acid (20% w/w, based on EFYS-PVA content) were added as plasticizers, and the mixture was stirred at 95°C for 30 min. The resulting film-forming solution (FFS) was divided into portions, to which 0%, 2%, 3%, or 4% (w/w) CTE anthocyanins were added and labeled C, S1, S2, S3. After heating, stirring, and degassing, 25 g of each FFS was cast onto 90 mm glass petri dishes and dried at 25°C for 24 h. The dried films were conditioned in a desiccator for 5 days before analysis to ensure stable and consistent moisture content (MC) and coded as C (CTE/MEFYS/PVA), S1 (2% CTE/MEFYS/PVA), S2 (3% CTE/MEFYS/PVA), and S3 (4% CTE/MEFYS/PVA) MEFYS/PVA/CTE films.

## 2.5 | Characterization of Films

### 2.5.1 | Physical Properties

**2.5.1.1 | Thickness and MC.** Film thickness (FT) was measured using a digital micrometer (Mitutoyo, Japan). Five measurements were taken at different points on each film, and the mean value was recorded. MC was determined gravimetrically by recording the weight loss of film samples before and after drying in a hot-air oven at 100°C until a constant weight was reached, using the following equation:

$$\text{MC (\%)} = \frac{M_i - M_f}{M_i} \times 100 \quad (3)$$

where  $M_i$  is the initial weight of film, and  $M_f$  is the final weight of film.

**2.5.1.2 | Swelling Rate, Water Solubility, and Water Vapor Permeability (WVP).** Film swelling rate was measured by immersing dried MEFYS/PVA/CTE films in distilled water at 25°C for 24 h. After immersion, surface moisture was removed and the films were weighed. The swelling degree was calculated as

$$\text{SR (\%)} = (M_0 - M_a)/M_a \times 100 \quad (4)$$

where  $M_a$  is the initial weight of film, and  $M_0$  is the final weight of the film.

Film solubility (FS) was determined by re-drying the swollen films at 60°C for 24 h and reweighing. FS was calculated as

$$\text{Film solubility (\%)} = (M_0 - M_1)/M_0 \times 100 \quad (5)$$

where  $M_0$  is the initial weight of film, and  $M_1$  is the final weight of the film after drying.

WVP was determined using a modified gravimetric method. Film samples (2 × 2 cm<sup>2</sup>) were sealed over silica gel tubes and placed

in a desiccator containing deionized water. Samples were weighed daily for 7 days, and WVP was calculated as

$$\text{WVP} = \frac{(m_1 - m_2) \times \Delta p}{S \times t \times d} \quad (6)$$

where  $(m_1 - m_2)$  is the increased weight of the tube (g),  $d$  is the FT (m),  $t$  is the testing time (s),  $A$  is the permeation area of the film ( $\text{m}^2$ ), and  $\Delta p$  is the water vapor pressure (2339 Pa)

**2.5.1.3 | Color Parameter of Film.** Film color parameters ( $L^*$ ,  $a^*$ ,  $b^*$ ) were measured using a digital colorimeter (Hunterlab ColorFlex EZ Spectrophotometer). The total color difference ( $\Delta E$ ) was calculated.

## 2.5.2 | pH Sensitivity of the CTE and CTE-EFY Films

For pH sensitivity, 1 mL sample of CTE and  $1 \times 1 \text{ cm}^2$  of CTE-EFY films were immersed in 10 mL of buffer solutions with pH values ranging from 1 to 12. The absorbance spectra were recorded between 400 and 700 nm using a UV-visible spectrophotometer (UH5300 UV-Visible spectrophotometer, Hitachi, Tokyo, Japan).

## 2.5.3 | Mechanical Properties

The mechanical properties of the films were measured following ASTM D882, including tensile strength (TS), elongation at break (EAB), and Young's modulus (YM). Tests were conducted at  $30^\circ\text{C}$  and 68% relative humidity using a micro tensile tester equipped with a load transducer. Data were analyzed using Texture Expert software, and all tests were carried out in triplicate:

$$\text{TS (MPa)} = \frac{F}{A} \quad (7)$$

$$\text{EAB (\%)} = \frac{L}{L_0} \times 100 \quad (8)$$

$$\text{YM} = \frac{\text{TS}}{S} \quad (9)$$

where  $F$  is the maximum force applied at rupture,  $A$  is the cross-sectional area of the film,  $L$  is the elongated length of the film at the point of rupture,  $L_0$  is the original length of the film before stretching, TS is the tensile strength, and  $S$  is the strain ( $L/L_0$ ).

## 2.5.4 | Morphological Examination

**2.5.4.1 | Scanning Electron Microscopy.** The morphology of starch granules and the films was examined using an FE-SEM (Tescan Mira-3, Czech Republic) operated at 5 kV. Film samples ( $1 \times 1 \text{ cm}^2$ ) were mounted on aluminum stubs with double-sided carbon tape, sputter-coated with gold using a vacuum evaporator, and observed on both surface and cross-sectional views.

## 2.5.4.2 | Confocal Laser Scanning Microscopy (CLSM).

The microstructure of the films was observed using CLSM (Olympus FV4000, Japan). Imaging was based on the autofluorescence of the film matrix, with excitation at 405 nm using a DAPI laser channel. Images were captured at a depth of  $20 \mu\text{m}$  below the coverslip using a  $20\times$  objective lens.

## 2.5.5 | X-Ray Diffraction (XRD) Analysis

The crystalline structures of the films were characterized using XRD with a  $\text{Cu K}\alpha$  radiation source ( $\lambda = 1.5406 \text{ \AA}$ ). The instrument operated at 40 kV and 40 mA, scanning over a  $2\theta$  range of  $5^\circ$ – $50^\circ$ .

## 2.5.6 | Fourier Transform Infrared Spectroscopy (FTIR)

FTIR spectra of the films were recorded at room temperature using an FTIR spectrometer (ALPHA II, Bruker, Germany) in the range of  $500$ – $4000 \text{ cm}^{-1}$ .

## 2.5.7 | Thermo-Gravimetric Analysis (TGA)

Thermal stability of the films was evaluated using a thermogravimetric analyzer (DTG-60 Shimadzu, Japan). Samples were heated from room temperature to  $400^\circ\text{C}$  at a heating rate of  $10^\circ\text{C}/\text{min}$  under a nitrogen atmosphere.

## 2.5.8 | Antioxidant Property

The antioxidant activity of the film samples was evaluated using DPPH and ABTS assays following the method of Wu et al. (2024). For the DPPH assay, 50 mg of each film sample was combined with a DPPH methanol solution. The mixture was then incubated in the dark for 30 min. Following incubation, the absorbance was measured at 517 nm, and the DPPH radical scavenging activity was calculated. For the ABTS assay, a working solution was prepared by reacting ABTS with potassium persulfate, and this mixture was allowed to stand in the dark overnight to ensure complete radical formation. The absorbance of the samples was subsequently recorded at 734 nm, and the antioxidant activity was calculated using the appropriate formula.

## 2.5.9 | Antimicrobial Activity

The antibacterial activity of the films was assessed using the agar well diffusion method described by Li et al. (2024). Two representative bacterial strains were used: *Staphylococcus aureus* (Gram-positive) and *Escherichia coli* (Gram-negative). Bacterial suspensions were adjusted to the 0.5 McFarland standard ( $\sim 1.5 \times 10^8 \text{ CFU/mL}$ ). A  $100 \mu\text{L}$  aliquot of each suspension was spread on Luria-Bertani agar plates, and wells (10 mm diameter) were filled with film discs. Ampicillin served as the positive control. The plates were incubated at  $37^\circ\text{C}$  for 24 h, after



which inhibition zone diameters were measured. All tests were performed in triplicate.

### 2.5.10 | Application for Monitoring Chicken Freshness

The effectiveness of the pH-sensitive indicator films for monitoring chicken meat freshness was tested following method by Kumar et al. (2024). Film indicators were attached to the lids of petri dishes containing 20.0 g chicken meat samples, which were stored at 4°C (refrigerator) and 37°C (incubator). During storage, the pH of the meat and the corresponding color changes of the films were periodically recorded. Microbiological quality was assessed by determining the total viable count (TVC). For this, 1.0 g of raw minced chicken meat was homogenized in sterile saline, serially diluted, and plated on plate count agar (PCA). The plates were inoculated at 30°C for 48 h, and TVC values were expressed as log colony-forming units per gram (log CFU/g). Chemical indicators of freshness were also measured. Total volatile basic nitrogen (TVB-N) was determined using a Kjeldahl apparatus (Borosil KDI040). Lipid oxidation was analyzed by the TBARS assay following Wang et al. (2021), with values calculated using an absorbance coefficient of 7.8 and expressed as mg malondialdehyde eq/kg sample. The pH of the supernatant was measured using a digital pH meter (Mettler Toledo). Chromatic properties of the films were quantified using a digital colorimeter (HunterLab ColorFlex EZ), and the color difference ( $\Delta E$ ) values were calculated to assess visual changes.

### 2.5.11 | Film Stability

Film stability was evaluated by monitoring color changes during storage. Samples were stored under controlled conditions (25°C and 35% RH) for 6 months. Color measurements and photographs were taken at 1, 3, and 6 months to assess stability over time.

## 2.6 | Statistical Analysis

All experiments were conducted in triplicate. Data were analyzed using SPSS software. One-way analysis of variance (ANOVA) followed by Duncan's multiple range test (DMRT) was used to determine significant differences among means at the  $p < 0.05$  level.

## 3 | Results and Discussion

### 3.1 | Total Starch Content and Amylose Content

A white, refined starch powder was successfully obtained through physical separation and drying, with a yield of 24% on a dry basis after dual-modification. Analysis revealed that the total starch content of isolated EFY starch was 79.45%, indicating high purity. The amylose content was 24.18%, whereas amylopectin accounted for 75.82%. This composition is consistent with the study reported by Iqbal et al. (2024) and plays a crucial role in film-forming ability and functional properties. Lower amylose content contributes

to reduced retrogradation, improved flexibility, and enhanced transparency but may decrease film strength. Therefore, blending with other polymers such as PVA and plasticizers is necessary to improve mechanical properties (Nagar et al. 2020). Thermal analysis showed an onset temperature ( $T_o$ ) of 65.62°C, peak temperature ( $T_p$ ) of 88.43°C, conclusion temperature ( $T_c$ ) of 100.21°C, and gelatinization enthalpy ( $\Delta H$ ) of 22.88 J/g. The relatively high  $\Delta H$  suggests a more ordered crystalline structure, that requires greater energy for gelatinization. Such characteristic thermal behaviour is consistent with a comparatively rigid starch matrix, which is advantageous for film formulation. Overall, the dual-modified EFY starch (MEFYS) exhibits promising characteristics, making it a suitable substrate for the developing, flexible, pH-responsive packaging films.

### 3.2 | Starch Particle Size and Morphology

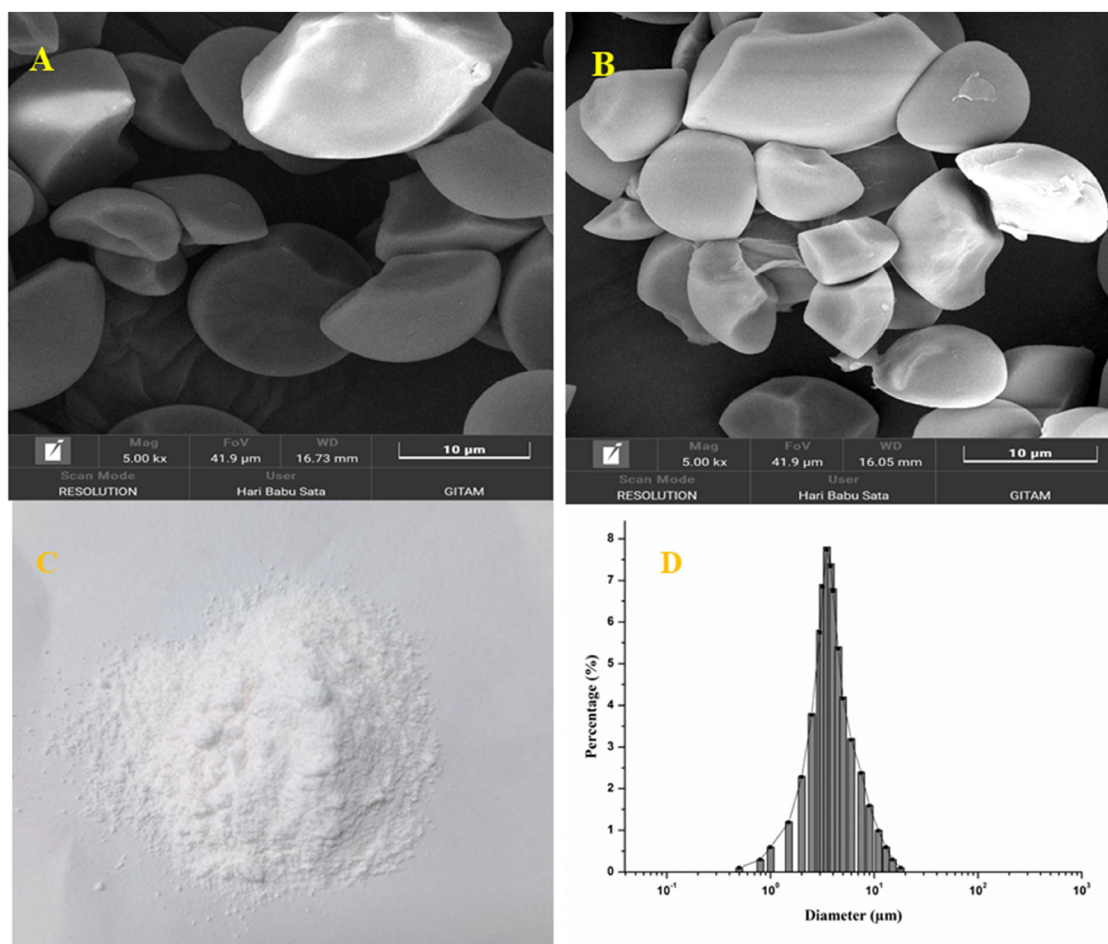
Morphological analysis of native and dual-modified EFY starches using FE-SEM (Figure 1A,B) revealed typical tuber-derived features, with smooth, uniform granules exhibiting oval to elliptical shapes. After treatment, granules appeared mainly oval or spherical with smooth surfaces. The final white, starch powder (Figure 1C) indicated effective drying and pigment removal, supporting its suitability for edible films and pharmaceutical applications. Laser diffraction analysis (Figure 1D) showed a median particle diameter of 26.30  $\mu\text{m}$  and mean diameter of 27.20  $\mu\text{m}$ .

### 3.3 | Anthocyanin Content of CTE

To ensure that spectrophotometric readings remained within the linear range, crude and its lyophilized CTE extracts were diluted 15- and 7-fold, respectively. The TAC of the solid CTE was 1.70 mg/g, whereas the dried butterfly pea flower powder yielded a TAC of 93.90 mg/100 g. Although these values are slightly lower than those reported for ultrasound-assisted extraction (Netravati et al. 2022), the TAC obtained in this study exceeded most previously published results. This comparatively higher yield may be related to the extent of 12 h, which likely enhanced pigment recovery.

### 3.4 | LC-MS/MS (Q-TRAP) Analysis of CTE

High-resolution LC-MS/MS QTRAP analysis of CTE identified 12 distinct metabolites (Table 1 and Figure S1), spanning multiple phytochemical classes, including carboxylic acids, flavonoids, flavonoid glycosides, phenolic acids, aldehydes, methyl esters, and alkaloids. Among these, key flavonoids, such as rutin, quercetin, delphinidin, and isoquercetin, were confirmed. Quantification reported in our recent study (Prakruthi et al. 2025) demonstrated that *C. ternatea* contains appreciable levels of ferulic acid (125.19  $\mu\text{g}/100\text{ mg}$ ), rutin (19.20  $\mu\text{g}/100\text{ mg}$ ), ellagic acid (9.63  $\mu\text{g}/100\text{ mg}$ ), myricetin (3.20  $\mu\text{g}/100\text{ mg}$ ), quercetin (5.41  $\mu\text{g}/100\text{ mg}$ ), apigenin 7-glucoside (0.85  $\mu\text{g}/100\text{ mg}$ ), and coumarin (1.63  $\mu\text{g}/100\text{ mg}$ ), with smaller quantities of gallic acid and protocatechuic acid also detected, which provides a



**FIGURE 1** | (A) Microstructure of native elephant foot yam (EFY) starch, (B) dual-modified EFY starch, (C) dual-modified EFY starch powder, and (D) particle size distribution of EFYS.

strong basis for justifying their incorporation and expected pH sensitivity. Chromatographic separation revealed that smaller carboxylic acids eluted earlier, whereas larger polyphenols and glycosides appeared within the 6–7 min retention time. Alkaloids (e.g., qingyangshengenin) and aldehydes (e.g., 2-methoxycinnamaldehyde) were detected at later retention times. Carboxylic acids, such as DL-malic acid and L-2-hydroxyglutaric acid, were predominantly identified in the negative ionization mode. These metabolites are often associated with redox homeostasis. Flavonoids, such as delphinidin, quercetin, and oxyresveratrol, are widely reported for their antioxidant, anti-inflammatory, and anticancer activities (Thuy et al. 2021). Similarly, phenolic acids and aldehydes contribute to the antioxidant capacity of the extract, while also offering antimicrobial and anti-inflammatory activities. The detection of qingyangshengenin, a pharmacologically active alkaloid, further broadens the functional profile of CTE reported in the literature. Overall, the metabolomic profile underscores the richness of CTE in bioactive secondary metabolites, with a predominance of antioxidant-related compounds supporting its potential for functional food and biomedical applications (Escher et al. 2020). The use of LC–MS/MS provided high-confidence identification, especially for structurally similar or isobaric compounds.

### 3.5 | Characterization of Films

#### 3.5.1 | Physical Properties

Food packaging films require adequate physical properties to ensure effective protection of food products. These characteristics include thickness, MC, solubility, swelling, and WVP. Incorporation of CTE significantly influenced all these properties ( $p < 0.05$ ), highlighting strong interactions between its bioactive compounds and the film matrix (Table 2).

#### 3.5.2 | Thickness and MC

As shown in Table 2, FT increased significantly with higher CTE concentrations, ranging from 0.12 mm in the control (C) to 0.16 mm in S3 (4% CTE). This increase suggests that polyphenol–matrix interactions promoted the formation of a denser, more structured film network, likely through hydrogen bonding and CTE-induced matrix expansion. MC decreased consistently from 16.2% (C) to 12.6% (S3). This reduction is attributed to phenolic compounds binding to starch molecules via intermolecular hydrogen bonding, thereby limiting the availability of hydrophilic

**TABLE 1** | Metabolites of *Clitoria ternatea* extract (CTE) identified in LC–MS/MS QTRAP.

| Pk no. | t <sub>R</sub> | Area       | Adduct/ Charge | Molecular formula                               | Precursor mass | Measured mass | Expected compound         | Major class         | References           |
|--------|----------------|------------|----------------|-------------------------------------------------|----------------|---------------|---------------------------|---------------------|----------------------|
| 1      | 2.11           | 4.39E + 08 | [M – H]–       | C <sub>4</sub> H <sub>6</sub> O <sub>5</sub>    | 133.35         | 133.3439      | DL-Malic acid             | Carboxylic acid     | Escher et al. (2020) |
| 2      | 2.14           | 3.85E + 07 | [M – H] –      | C <sub>5</sub> H <sub>8</sub> O <sub>5</sub>    | 147.6          | 147.6119      | L-2-Hydroxyglutaric acid  | Carboxylic acid     |                      |
| 3      | 5.99           | 4.14E + 07 | [M + H]+       | C <sub>13</sub> H <sub>22</sub> O <sub>3</sub>  | 227.301        | 227.3303      | Methyl dihydrojasmonate   | Methyl ester        |                      |
| 4      | 6.37           | 3.06E + 08 | [M + H]+       | C <sub>15</sub> H <sub>11</sub> O <sub>7</sub>  | 303.1          | 303.0957      | Delphinidin cation        | Flavonoid           |                      |
| 5      | 6.37           | 2.21E + 08 | [M + H]+       | C <sub>27</sub> H <sub>30</sub> O <sub>16</sub> | 611.487        | 611.4882      | Rutin                     | Flavonoid glycoside |                      |
| 6      | 6.44           | 2.08E + 08 | [M + H]+       | C <sub>15</sub> H <sub>10</sub> O <sub>7</sub>  | 303.088        | 303.0879      | Quercetin                 | Flavonoid           |                      |
| 7      | 6.56           | 4.00E + 08 | [M + H]+       | C <sub>27</sub> H <sub>30</sub> O <sub>15</sub> | 595.6          | 595.6172      | Kaempferol 3-O-rutinoside | Flavonoid glycoside |                      |
| 8      | 6.65           | 1.83E + 08 | [M + H]+       | C <sub>21</sub> H <sub>20</sub> O <sub>12</sub> | 465.533        | 465.5253      | Isoquercetin              | Flavonoid           |                      |
| 9      | 10.05          | 2.33E + 08 | [M – H] –      | C <sub>11</sub> H <sub>12</sub> O <sub>4</sub>  | 207.421        | 207.425       | Ferulic acid methyl ester | Phenolic acid       |                      |
| 10     | 12             | 6.94E + 07 | [M + H]+       | C <sub>10</sub> H <sub>10</sub> O <sub>2</sub>  | 163.323        | 163.323       | 2-Methoxycinnamaldehyde   | Aldehyde            |                      |
| 11     | 12.97          | 1.08E + 08 | [M + H]+       | C <sub>49</sub> H <sub>72</sub> O <sub>17</sub> | 482.992        | 482.9974      | Qingyangshengenin         | Alkaloid            |                      |
| 12     | 14.05          | 5.77E + 07 | [M – H] –      | C <sub>14</sub> H <sub>12</sub> O <sub>4</sub>  | 243.337        | 243.3391      | Oxyresveratrol            | Flavonoid           |                      |

**TABLE 2** | Physical properties of the films.

| Film type | Thickness (mm)           | Moisture (%)            | Film solubility (%)     | Swelling rate (%)        | WVP (10 <sup>–10</sup> g/m/s/Pa) |
|-----------|--------------------------|-------------------------|-------------------------|--------------------------|----------------------------------|
| C-0%      | 0.12 ± 0.01 <sup>d</sup> | 16.2 ± 0.5 <sup>a</sup> | 27.5 ± 1.2 <sup>a</sup> | 21.5 ± 1.3 <sup>a</sup>  | 1.77 ± 0.09 <sup>a</sup>         |
| S1-2%     | 0.14 ± 0.02 <sup>c</sup> | 14.5 ± 0.6 <sup>b</sup> | 24.8 ± 1.0 <sup>b</sup> | 19.8 ± 1.6 <sup>b</sup>  | 1.52 ± 0.13 <sup>b</sup>         |
| S2-3%     | 0.15 ± 0.02 <sup>b</sup> | 13.8 ± 0.7 <sup>c</sup> | 23.5 ± 1.1 <sup>c</sup> | 18.2 ± 0.52 <sup>c</sup> | 1.48 ± 0.18 <sup>c</sup>         |
| S3-4%     | 0.16 ± 0.01 <sup>a</sup> | 12.6 ± 0.4 <sup>d</sup> | 21.0 ± 0.8 <sup>d</sup> | 16.9 ± 0.51 <sup>d</sup> | 1.46 ± 0.18 <sup>d</sup>         |

Note: Data are presented as the mean ± SD. Different superscript letters in the same column indicate significant differences ( $p < 0.05$ ). Film codes: C (0% CTE), S1 (2% CTE), S2 (3% CTE), and S3 (4% CTE) MEFY starch–PVA films.

sites and increasing the hydrophobicity of the matrix. A lower MC enhances film stability and shelf life under humid storage conditions (Wu et al. 2024).

crosslinking and reduced free volume within the film matrix. Such reduced swelling favors applications in moisture-sensitive packaging and controlled-release systems.

### 3.5.3 | Swelling Rate and FS

FS decreased from 27.5% in the control to 21.0% in S3. At lower extract concentrations (S1 and S2), solubility showed only a slight reduction, likely due to initial polymer disruption. However, at higher CTE (S3), stronger polyphenol-polymer interactions and cross-linking enhanced water resistance, thereby lowering solubility. Similar trends have been reported in starch–polyphenol systems, where phenolics act as natural cross-linkers that improve water stability (Zhu et al. 2021). The swelling rate decreased from 21.5% in the control to 16.9% in S3, indicating the formation of a tighter polymeric network with reduced water uptake capacity. This effect can be attributed to flavonoid-induced

### 3.5.4 | Water Vapor Permeability

WVP decreased from  $1.77 \times 10^{-10}$  g/m/s/Pa (C) to  $1.46 \times 10^{-10}$  g/m/s/Pa (S3), demonstrating enhanced barrier properties. This improvement is associated with a more compact matrix and reduced porosity. Although very high CTE levels could potentially introduce micro-voids, the tested formulations maintained strong barrier integrity. Overall, Efy/PVA/CTE composite films exhibited superior water resistance and barrier performance compared with many conventional starch-based films, highlighting their suitability for food packaging applications (Zhu et al. 2021).

**TABLE 3** | The color parameters of films with different *Clitoria ternatea* extract (CTE) contents.

| Film type | $L^*$             | $a^*$             | $b^*$              | $\Delta E$        |
|-----------|-------------------|-------------------|--------------------|-------------------|
| C-0%      | $88.6 \pm 0.43^a$ | $3.2 \pm 0.15^d$  | $-4.9 \pm 0.26^a$  | $1.80 \pm 0.10^d$ |
| S1-2%     | $60.4 \pm 0.30^b$ | $6.4 \pm 0.15^c$  | $-22.5 \pm 0.30^b$ | $12.2 \pm 0.20^c$ |
| S2-3%     | $55.8 \pm 0.43^c$ | $9.7 \pm 0.17^b$  | $-26.8 \pm 0.26^c$ | $14.6 \pm 0.20^b$ |
| S3-4%     | $35.8 \pm 0.70^d$ | $14.5 \pm 0.30^a$ | $-29.2 \pm 0.35^d$ | $17.5 \pm 0.30^a$ |

Note: Data are presented as the mean  $\pm$  SD. Different superscript letters in the same column indicate significant differences ( $p < 0.05$ ). Film codes: C (0% CTE), S1 (2% CTE), S2 (3% CTE), and S3 (4% CTE) MEFY starch-PVA films.

### 3.5.5 | Color Parameter of Film

The incorporation of CTE significantly altered the color characteristics of the films, as shown in Table 3 and Figure S2. With increasing CTE concentration (0%–4%), marked shifts were observed in the CIE  $Lab^*$  color space. The  $L^*$  value (lightness) decreased sharply from 88.6 in the control film to 35.8 in S3 film (4% CTE), indicating a darker film appearance due to pigmented polyphenols and flavonoids, such as delphinidin and quercetin. The  $a^*$  value (red/green axis) increased from 3.2 in the control to 14.5 in S3, reflecting the red-violet anthocyanins, particularly delphinidin. In contrast, the  $b^*$  value (yellow/blue) decreased progressively from  $-4.90$  to  $-29.20$ , signifying an intensified shift toward blue tones, consistent with the presence of blue-purple flavonoids. The overall color difference ( $\Delta E$ ) increased significantly with higher CTE levels, with films containing 3%–4% extract showing visibly distinct appearances compared with the control. Such pronounced color changes enhance the films' potential for functional labeling, product differentiation, and smart packaging applications, where visual cues can serve as indicators of composition or freshness (Wang et al. 2023).

### 3.6 | pH Sensitivity of the CTE and CTE-EFY Films

CTE exhibited distinct pH-dependent color transitions, shifting from a pink hue in acidic conditions to green-yellow color in alkaline environments (Figure 2). UV-visible spectroscopy confirmed these visual changes, showing a bathochromic shift in the maximum absorption wavelength from  $\sim 520$  to  $540$  nm (flavylium cation, pink color) to around  $625$  nm (quinoidal base/chalcone forms). This shift reflects the reversible protonation-deprotonation of anthocyanins, which exist in different ionic forms across the pH spectrum.

Both CTE and EFY/PVA/CTE films displayed strong and reversible color responses over a wide pH range, confirming that film incorporation did not diminish sensitivity. The observed visual transitions are governed by the multistate equilibrium of anthocyanins and related flavonoids, enabling the films to act as natural colorimetric pH indicators. Importantly, CTE showed greater stability than many conventional natural dyes, further enhancing its suitability for long-term use. These properties highlight the film's potential in intelligent packaging, where real-time visual cues can signal food freshness, spoilage, or package integrity. The behavior is consistent with findings from other anthocyanin-based systems, such as cassava starch films and *Lycium ruthenicum* anthocyanin (Qin et al. 2019).

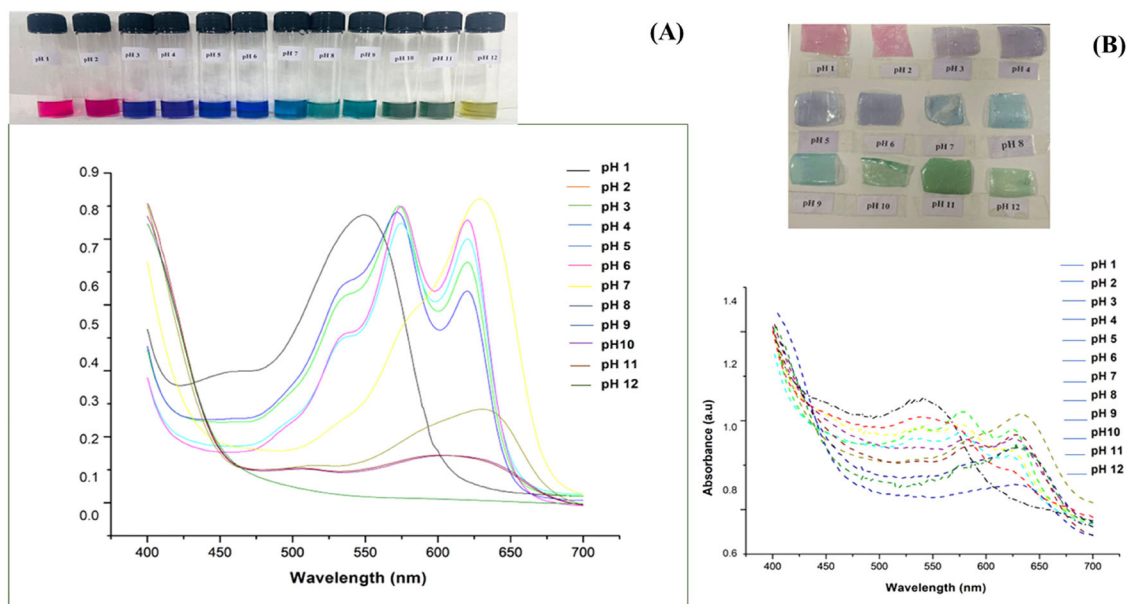
### 3.7 | Mechanical Properties of Films

Table 4 summarizes the mechanical properties of EFY/PVA/CTE films at different CTE concentrations. TS increased significantly with CTE incorporation, reaching a maximum of  $16.36$  MPa at 3% CTE film (S2), compared to  $13.86$  MPa in the control (C). This enhancement is attributed to strong interfacial interactions and uniform dispersion of CTE, which reinforced the polymer matrix. Similarly, YM rose with CTE addition, with S1 and S3 showing higher stiffness ( $71.40$  and  $69.26$  MPa, respectively), indicating enhanced rigidity. EAB, however, showed a nonlinear trend: It initially increased to  $24.20\%$  at S2, suggesting greater flexibility, but declined to  $21.30\%$  at S3, reflecting reduced chain mobility and filler-induced rigidity at higher loadings. Overall, the S2 formulation (3% CTE) achieved the best balance between strength, stiffness, and flexibility, making it the most promising among the tested films. In contrast, further increasing CTE content (S3) compromised ductility without yielding notable improvements in strength, emphasizing the importance of optimizing active compound concentration in composite film design. Similar concentration-dependent effects on mechanical performance have been reported in starch- and PVA-based pH-intelligent films (Wang et al. 2023; Zhu et al. 2021).

### 3.8 | Morphological Characteristics

The morphological and compositional characteristics of EFY/PVA/CTE composite films were evaluated using FE-SEM and CLSM (Figure 3). As shown in surface scanning electron microscopy (SEM) images (Figure 3A), the control film (C) exhibited a relatively smooth surface with only minor pores, suggesting uniform blending of EFYS and PVA in the absence of polyphenol fillers. In S1, the addition of CTE increased surface roughness and produced micro-voids, indicative of partial matrix disruption. S2 displayed a smoother, denser surface with minimal defects, highlighting better filler dispersion and polymer-filler compatibility. By contrast, S3 showed a coarse surface with visible particle agglomerates, likely associated with oversaturation and poor CTE dispersion. Cross-sectional SEM images (Figure 3B) revealed that the control film has loosely packed, porous internal structure. S1 showed partial integration of CTE with discontinuities at the filler-matrix interface. S2 exhibited the most compact and continuous internal morphology, suggesting optimal filler-matrix interaction. In S3, disrupted layers, voids, and stress cracks were evident, indicating compromised structural integrity due to CTE agglomeration.





**FIGURE 2** | The colors and UV-vis spectra of CTE (A) and EFY/PVA/CTE films (B) in buffers with different pH values.

**TABLE 4** | Mechanical properties of the films.

| CTE content | TS (MPa)                  | EAB (%)                  | YM (MPa)                 |
|-------------|---------------------------|--------------------------|--------------------------|
| C-0%        | 13.86 ± 0.80 <sup>d</sup> | 19.7 ± 0.8 <sup>d</sup>  | 70.36 ± 2.0 <sup>b</sup> |
| S1-2%       | 16.13 ± 0.20 <sup>b</sup> | 22.6 ± 1.0 <sup>b</sup>  | 71.40 ± 2.5 <sup>a</sup> |
| S2-3%       | 16.36 ± 0.35 <sup>a</sup> | 24.20 ± 1.2 <sup>a</sup> | 67.60 ± 3.0 <sup>d</sup> |
| S3-4%       | 14.75 ± 0.36 <sup>c</sup> | 21.30 ± 0.6 <sup>c</sup> | 69.26 ± 2.2 <sup>c</sup> |

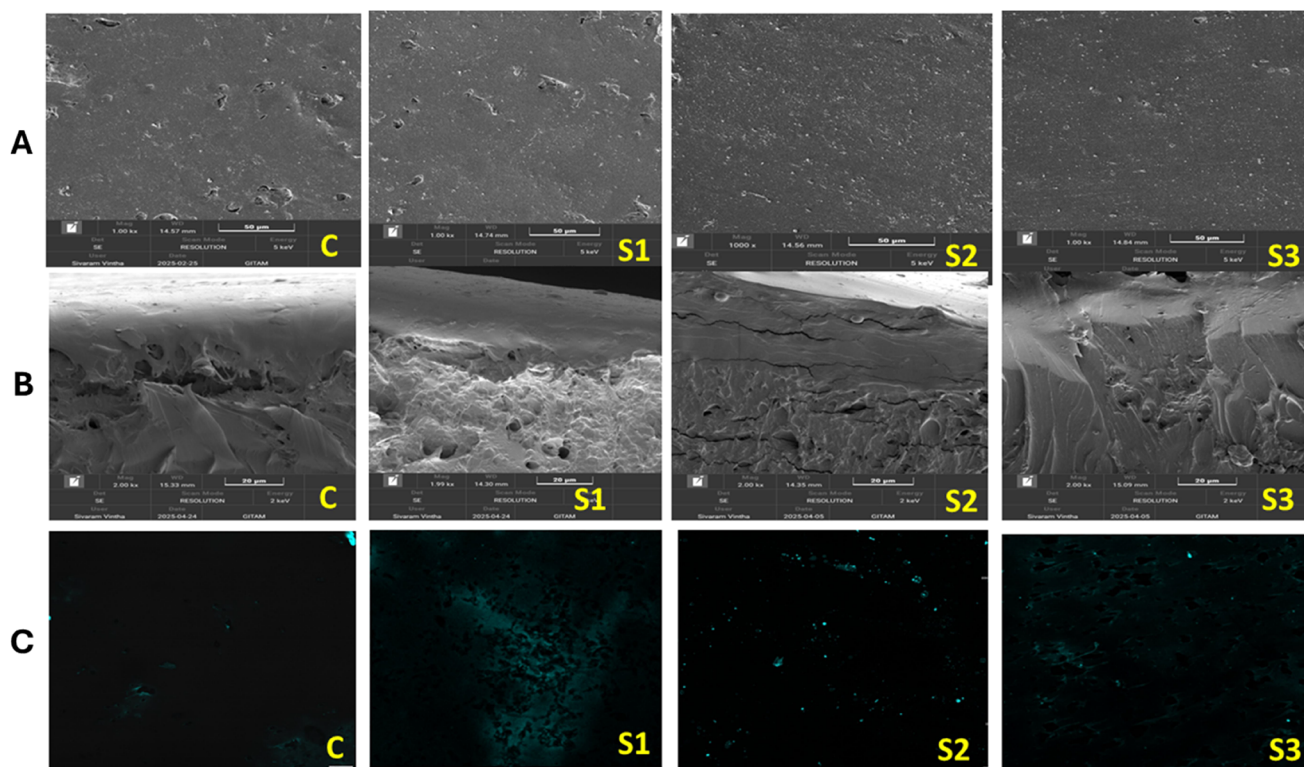
Note: Data are presented as the mean ± SD. Different superscript letters in the same column indicate significant differences ( $p < 0.05$ ). Film codes: C (0% CTE), S1 (2% CTE), S2 (3% CTE), and S3 (4% CTE) MEFY starch-PVA films.

CLSM imaging (Figure 3C) revealed the fluorescence differences in CTE dispersion. The control film showed negligible fluorescence, whereas S1 exhibited moderate but uneven distribution. S2 revealed strong, uniform fluorescence, consistent with well-dispersed CTE. In contrast, S3 displayed intense but irregular fluorescence, confirming CTE aggregation. Overall, the S2 formulation proved optimal, combining a compact morphology, strong filler-matrix integration, and uniform CTE distribution. These features jointly enhance film performance and align with findings reported in bilayer smart films (Wu and Li 2023).

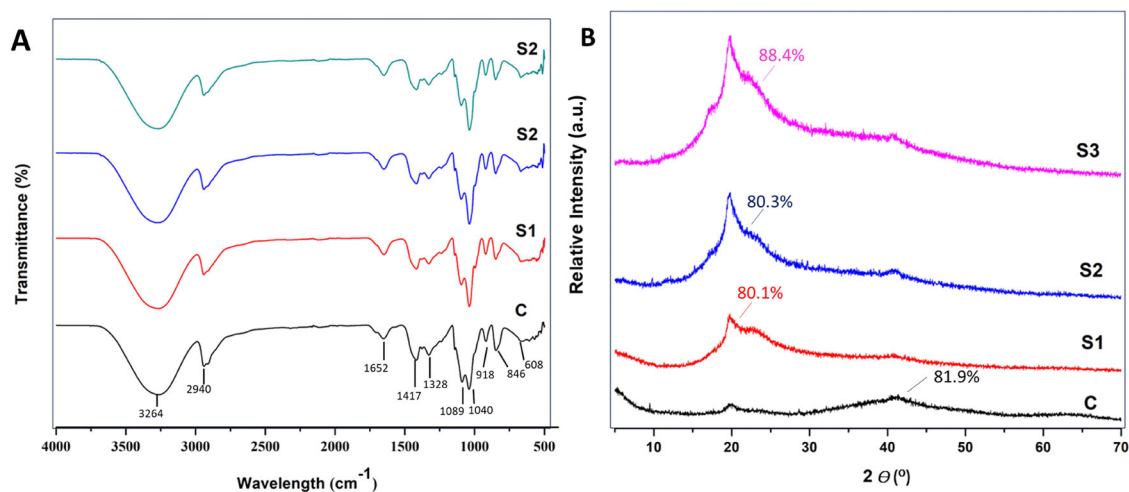
### 3.9 | XRD Analysis

The XRD patterns, presented in Figure 4A, offer valuable insights into the crystallinity and molecular ordering within the various film matrices. The control film (C) exhibited a broad diffraction halo between  $2\theta = 19^\circ$  and  $21^\circ$ , characteristic of semicrystalline materials, such as PVA-starch systems. Crystallinity (%) values reported (C: 18.1%; S1: 19.9%; S2: 19.7%; S3: 11.6%) were calculated by deconvolution of the crystalline and amorphous contributions in the XRD profiles and are plotted in Figure 4A alongside the relative intensity traces. With the incorporation of CTE, crystallinity initially increased from 18.1%

in C to 19.9% in S1, accompanied by a slightly sharper and more intense halo, indicating improved molecular alignment or ordering promoted by interactions between CTE components and the polymer chains. However, further addition of CTE resulted in reduced crystallinity, decreasing to 19.7% in S2 and 11.6% in S3. Correspondingly, the intensity of the broad peak became sharper in S1 suggesting an enhanced ordering or molecular alignment within the film structure, likely due to interactions between the CTE molecules and the polymer chains. However, the reduced crystallinity of S3 points to a disruption of chain packing at higher CTE loading. This reduction reflects disruption of polymer chain packing at higher extract concentrations, likely due to excessive polyphenolic interference or flavonoid-polymer intercalation. These modifications suggest that moderate CTE incorporation promotes formation of ordered, crystalline-like domains, whereas higher loading disrupts structural regularity. These structural modifications could arise from polyphenolic stacking or flavonoid-PVA alignment, forming crystalline-like domains in moderate concentrations but weakening order at higher levels. The observed trend is consistent with corresponding changes in mechanical flexibility and swelling index, tensile behavior, and SEM, all of which support enhanced compactness at moderate CTE levels and reduced structural regularity at higher concentrations. This semicrystalline enhancement is particularly beneficial for films that require moderate rigidity, good



**FIGURE 3** | (A) Surface, (B) cross-sectional scanning electron microscope photographs; (C) CLSM images of EFY/PVA/CTE films. Film codes: C (0% CTE), S1 (2% CTE), S2 (3% CTE), and S3 (4% CTE) MEFY starch-PVA films.

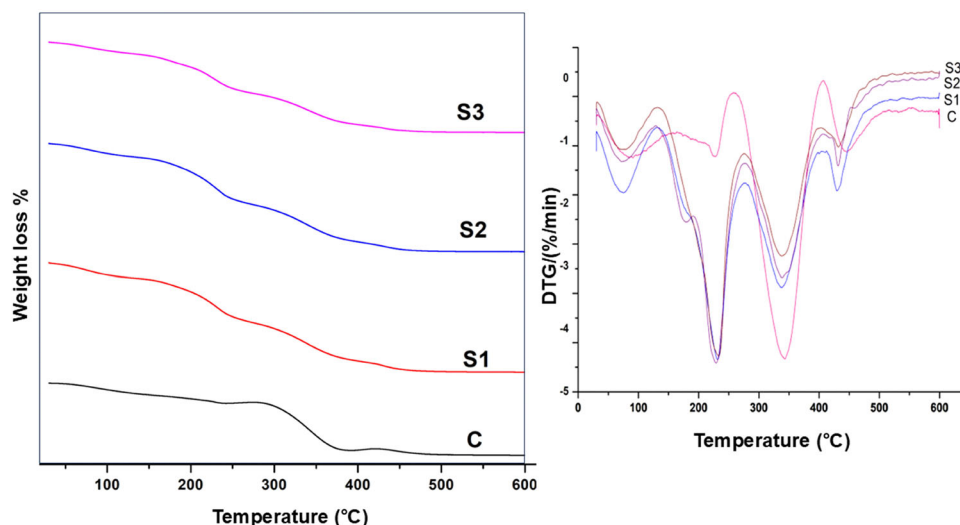


**FIGURE 4** | FTIR spectra (A) and x-ray diffraction patterns (B) of the films. Film codes: C (0% CTE), S1 (2% CTE), S2 (3% CTE), and S3 (4% CTE) MEFY starch-PVA films.

optical clarity, and a slower degradation rate. Overall, the XRD analysis confirms that CTE integrates effectively into the starch-PVA matrix and induces a structural reorganization, leading to an enhancement of crystalline features without evidence of phase separation. This observation is consistent with findings reported in previous studies on similar composite systems (Fitch-Vargas et al. 2016; Y. Wu and Li 2023).

### 3.10 | Fourier Transform Infrared Spectroscopy (FTIR)

The FTIR spectra (Figure 4B) revealed characteristic absorption bands for both the control film (C) and the CTE-incorporated films (S1–S3), confirming interactions among starch, PVA, and bioactive components. All films exhibit a broad O–H stretching



**FIGURE 5** | Thermo-gravimetric analysis curves of the films. Film codes: C (0% CTE), S1 (2% CTE), S2 (3% CTE), and S3 (4% CTE) MEFY starch-PVA films. DTG, derivative thermogravimetric.

band around  $\sim 3260\text{--}3300\text{ cm}^{-1}$ , attributed to extensive hydrogen bonding. The band becomes broader and slightly more intense in S1–S3, indicating strengthened hydrogen bonding due to phenolic hydroxyl groups from CTE. The C–H stretching vibration near  $\sim 2940\text{ cm}^{-1}$  is present across all samples, corresponding to aliphatic  $\text{--CH}_2$  groups of the starch-PVA backbone. Peaks around  $\sim 1650\text{ cm}^{-1}$  (C=O stretching) and  $\sim 1415\text{--}1328\text{ cm}^{-1}$  (C–H bending) become more pronounced in extract-loaded films, which is consistent with aromatic rings and carboxyl functionalities present in CTE flavonoids. In the fingerprint region ( $1200\text{--}800\text{ cm}^{-1}$ ), characteristic C–O–C, C–OH, and glycosidic vibrations appear in all films. Notably, shifts and increased complexity in S2 and S3 suggest in peaks near 1089, 1040, 918, 846, and  $608\text{ cm}^{-1}$  suggest physical entrapment and non-covalent interactions between CTE constituents and the polymer network. These changes support successful incorporation of CTE and the formation of hydrogen-bond-driven interactions, which likely contribute to improved barrier functionality and bioactivity of the films. Overall, the FTIR findings validate the presence of phenolic and flavonoid moieties and their interaction with starch and PVA, in agreement with earlier reports (Li et al. 2022).

### 3.11 | Thermo-Gravimetric Analysis

The thermal degradation behavior of EFY/PVA/CTE films was analyzed using TGA and derivative thermogravimetric (DTG) curves to evaluate weight loss patterns, thermal stability, and the effect of CTE incorporation on degradation profiles. The TGA curves in Figure 5 revealed multistep weight loss behavior in all film samples, which is typical of polymeric bio-composites. The major degradation phase ( $200\text{--}400^\circ\text{C}$ ) is attributed to the thermal decomposition of the polymer backbone (starch and PVA) and the bioactive compounds in CTE. A residual weight plateau is observed beyond  $450^\circ\text{C}$ , suggesting the formation of thermally stable char residues. Films containing with CTE (S1–S3) exhibited higher total weight loss, particularly at higher concentrations (S2 and S3), reflecting the thermal contribution of phytochemical components. The DTG curves provided further

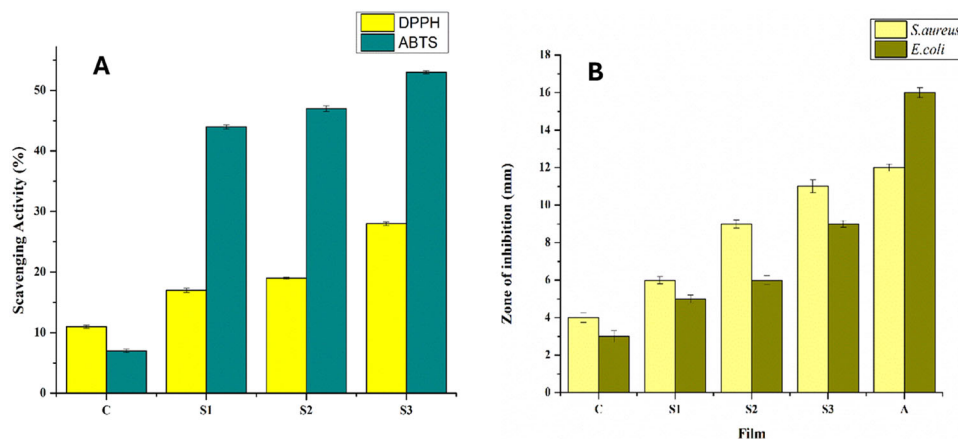
insight into the rate of decomposition, showing two to three main peaks corresponding to distinct degradation stages. The first major peak ( $\sim 250\text{--}300^\circ\text{C}$ ) is sharper in the control, indicating faster degradation, while the second degradation event ( $\sim 350\text{--}400^\circ\text{C}$ ) is more pronounced in S2 and S3, likely associated with the degradation of thermally sensitive flavonoids. The broadened peaks and delayed onset in films with higher CTE content suggest that CTE incorporation modifies the thermal decomposition pathway, possibly due to hydrogen bonding and network stabilization by phenolic groups. Overall, CTE incorporation alters thermal degradation, increasing total weight loss while promoting multiphase, slower decomposition, indicating improved structural interactions and thermal stability (Zhang et al. 2020).

### 3.12 | Antioxidant Property

Polyphenols, known for donating electrons or hydrogen atoms via their phenolic hydroxyl groups, exhibit potent antioxidant activity, which is crucial for mitigating oxidative degradation and enhancing food preservation during transport and storage. In this study, the CTE-loaded films demonstrated a significantly higher free radical scavenging activity, against both DPPH and ABTS radicals, compared to pure EFY starch films (Figure 6A). The DPPH radical scavenging capacity (17%–28%) of the CTE-loaded films was lower than their ABTS activity (44%–53%), likely due to limited solubility of certain CTE components in the DPPH assay solvent. Nonetheless, these results indicate that CTE is an effective polyphenol source, capable of donating hydrogen and neutralizing free radicals, highlighting its potential as a protective agent in active food packaging (Yong et al. 2019).

### 3.13 | Antimicrobial Property

The antimicrobial activity of the EFY starch films incorporating anthocyanins from CTE was evaluated against two common food-borne pathogens: *S. aureus* and *E. coli*. As shown in Figure 6B and



**FIGURE 6** | Antioxidant activity (A) and antimicrobial activity (B) of EFYS/PVA/CTE films. Film codes: C (0% CTE), S1 (2% CTE), S2 (3% CTE), and S3 (4% CTE) MEFYS films.

Figure S3A, the well diffusion assay confirmed clear inhibitory effects. The films exhibited concentration-dependent activity, with inhibition zones increasing from 4 mm (C) to 12 mm (S3) against *S. aureus* and from 3 mm (C) to 16 mm (S3) against *E. coli*. The enhanced antimicrobial effect is likely driven by delphinidin-based ternatins, the dominant anthocyanins in CTE, supported by flavanols such as rutin and quercetin (Jeyaraj et al. 2022). Consistent with Li et al. (2022), a stronger effect was observed against Gram-positive bacteria, as the absence of an outer membrane facilitates greater diffusion of bioactive compounds compared to Gram-negative strains. These findings highlight the potential of CTE-loaded films as antimicrobial packaging materials capable of inhibiting microbial growth thereby contributing to extending food shelf-life.

### 3.14 | Monitoring Chicken Freshness

Chicken, being protein-rich, is highly susceptible to spoilage caused by microbial and enzymatic activity. The accumulation of volatile compounds and an increase in TVC are widely recognized indicators of spoilage, with TVC values above 6 log CFU/mL generally signaling spoilage. In this study, TVC and pH measurements were correlated with color changes of the developed CTE-incorporated films to assess their ability to monitor chicken freshness (Figure 7A–D; Figure 3C). At 37°C, the TVC exceeded the spoilage threshold within 16 h, accompanied by a rise in pH to 6.75. Correspondingly, the intelligent films exhibited a distinct chromatic transition from blue-violet to blue, indicating the onset of spoilage. After 32 h, the chicken was severely spoiled (pH 7.25), and the film color shifted to blue-green. In contrast, samples stored at 4°C for 32 h remained fresh, with the TVC below the threshold and pH stable near 6.

Figure 7C shows temporal changes in TVB-N and TBARS, key indicators of protein degradation and lipid oxidation. At 0 h, TVB-N was 6.2 mg/100 g, and TBARS was below 0.5 mg MDA/kg, confirming product freshness. By 16 h at 37°C, TVB-N doubled to 14.7 mg/100 g, and TBARS rose to 1.2 mg MDA/kg, coinciding with initial spoilage. After 32 h, TVB-N reached 32.2 mg/100 g and TBARS peaked at 2.3 mg MDA/kg, confirming severe deterioration. These biochemical changes closely paralleled microbial

growth, pH elevation, and film color transitions. Although the films showed clear spoilage indication, color intensity decreased after 16 h, likely due to anthocyanin degradation under alkaline conditions. Similar fading trends have been reported in earlier studies (Yong et al. 2019; Zhang et al. 2020; Li et al. 2022). Despite this,  $\Delta E$  values at 37°C increased markedly with CTE content (from 38.89 in C to 66.13 in S3; Table 5), reflecting distinct color shifts from blue-violet to blue-green. At 4°C,  $\Delta E$  values remained low (18.96–23.20), consistent with minimal spoilage and stable freshness. Overall, these results demonstrate that CTE-loaded films function effectively as freshness indicators, offering real-time visual cues of spoilage through integrated biochemical and chromatic responses.

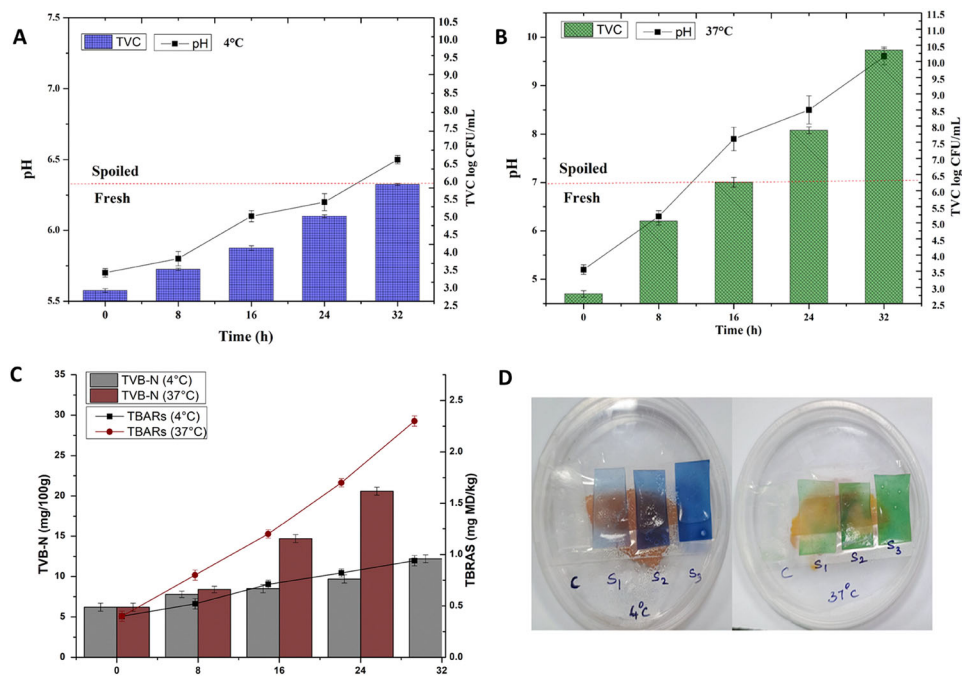
### 3.15 | Stability of the Films

The stability of intelligent packaging films was assessed on the basis of their color retention ability (Figure 8). Films with higher CTE content, particularly S3, exhibited a 68% color change after 6 months of storage, whereas S1 films with lower CTE content showed only a 25% change. This indicates that although higher CTE concentrations enhance functional properties, they compromise long-term color stability. Despite pigment loss, all films retained their pH-responsive color-shifting ability throughout the 6-month storage period. This preserved responsiveness confirms their continued effectiveness for real-time monitoring of food freshness, even with partial color degradation over time.

## 4 | Conclusion

This study successfully isolated and characterized EFY starch, an abundant and nontoxic agricultural byproduct, for the development of intelligent packaging film capable of indicating food freshness through pH-responsive color changes. The fabricated films demonstrated cost-effectiveness and potential for large-scale production. Incorporation of CTE significantly modified FT, ductility, antioxidant activity, and light barrier performance, thereby improving the functional performance of the films. Although minor mechanical parameters changed with CTE loading, MEFYS/PVA/CTE films maintained acceptable strength and



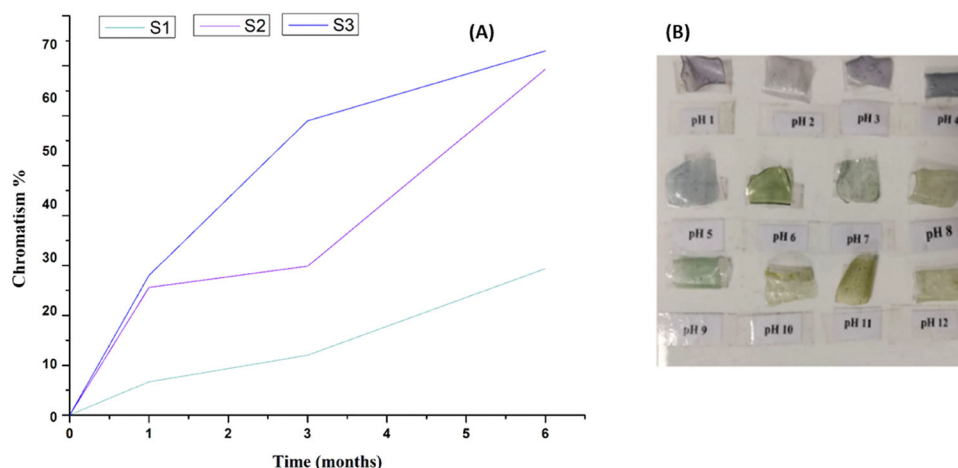


**FIGURE 7** | Changes of TVC and pH of chicken stored at 4°C (A) and 37°C (B) film at the given time; changes in TVB-N and TBARs of chicken samples stored at 4°C and 37°C (C) and real photos (D) during freshness monitoring. Film codes: C (0% CTE), S1 (2% CTE), S2 (3% CTE), and S3 (4% CTE) MEFYS-PVA films were arranged from left to right. TVB-N, total volatile basic nitrogen; TVC, total viable count.

**TABLE 5** | Color changes of the films after chicken spoilage.

| CTE content | $\Delta E$ (4°C)  | $\Delta E$ (37°C) |
|-------------|-------------------|-------------------|
| C-0%        | $18.96 \pm 1.0^d$ | $38.89 \pm 2.1^d$ |
| S1-2%       | $20.50 \pm 1.2^c$ | $50.35 \pm 1.4^c$ |
| S2-3%       | $22.10 \pm 1.5^b$ | $57.21 \pm 1.6^b$ |
| S3-4%       | $23.20 \pm 1.3^a$ | $66.13 \pm 2.5^a$ |

Note: Data are presented as the mean  $\pm$  SD. Different superscript letters in the same column indicate significant differences ( $p < 0.05$ ). Film codes: C (0% CTE), S1 (2% CTE), S2 (3% CTE), and S3 (4% CTE) MEFY starch-PVA films.



**FIGURE 8** | Shelf-life stability of films during storage—images of film (A) and chromatism values (B) of EFY-CTE films. Film codes: S1 (2% CTE), S2 (3% CTE), and S3 (4% CTE) MEFYS-PVA films.

barrier properties and improved functional properties compared to conventional starch films. Overall, the findings demonstrate that EFY starch and CTE can be effectively combined to produce sustainable intelligent packaging films with strong functional properties and reliable colorimetric responses. This work highlights a novel approach for valorizing underutilized EFYS and advances the development of smart packaging technologies.

## Author Contributions

**Likhitha Yadav Prkruthi:** conceptualization, methodology, software, data curation, investigation, writing – original draft, visualization, validation. **Hari Krishnan:** methodology, investigation, data curation, writing – original draft, validation, visualization. **Chagam Koteswara Reddy:** conceptualization, funding acquisition, project administration, supervision, resources, writing – review and editing. **Tapasya Kumari:** formal analysis, writing – review and editing, resources. **Ramachandran Vinayagam:** conceptualization, writing – review and editing, formal analysis.

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## Conflicts of Interest

The authors declare no conflicts of interest.

## Data Availability Statement

The data presented in this study are available on request from the corresponding authors.

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## Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supplementary Materials: jfds70758-sup-0001-figuresS1-S3.docx